Glossary

91 terms from every unit. Search, or jump by letter. Each entry links back to the unit where it is taught.

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

Unit 1: Your First Python Program

Assignment

Storing a value under a name using `=`. The right side is computed first.

Unit 1

Comment

Text after `#` that Python ignores. Used to explain intent to human readers.

Unit 1

Interpreter

The program that reads Python code and runs it. Python code runs line by line through the interpreter.

Unit 1

Keyword

A word reserved by the language (for example `if`, `for`, `True`). It cannot be used as a variable name.

Unit 1

NameError

The error raised when code uses a name that has not been assigned.

Unit 1

print()

A built-in function that writes its arguments to the output, separated by spaces, followed by a newline.

Unit 1

snake_case

Naming style with lowercase words joined by underscores, such as `total_price`. Python's convention for variables and functions.

Unit 1

Statement

One complete instruction the interpreter runs, such as `print("hi")` or `x = 3`.

Unit 1

SyntaxError

The error raised when code breaks the grammar of the language, such as a missing quotation mark or parenthesis.

Unit 1

Variable

A name that refers to a value stored in memory. The value can be changed by assigning a new one.

Unit 1

Unit 2: Data Types and Strings

bool

Boolean: the value `True` or `False`. Named for George Boole.

Unit 2

Casting

Converting a value from one type to another, for example `int("42")`.

Unit 2

Data type

A category of value (int, float, str, bool) that determines what operations are valid on it.

Unit 2

f-string

A string literal beginning with `f` that embeds expressions in braces, such as `f"{x}"`.

Unit 2

float

Floating-point number: a number with a decimal point. Stored approximately.

Unit 2

Immutable

A value that cannot be changed after it is created. Strings, integers, and tuples are immutable.

Unit 2

Index

The position of an item in a sequence. Counting starts at 0.

Unit 2

int

Integer: a whole number with no decimal point.

Unit 2

Method

A function that belongs to an object and is called with a dot, such as `name.upper()`.

Unit 2

Slice

A part of a sequence written `seq[start:stop]`. The stop position is excluded.

Unit 2

str

String: an immutable sequence of characters written in quotes.

Unit 2

type()

A built-in function that returns the type of a value, such as `<class 'int'>`.

Unit 2

Unit 3: Boolean Logic and Truth Tables

Boolean

A value that is either `True` or `False`.

Unit 3

Chained comparison

A comparison such as `1 < x < 10`, which Python evaluates as two comparisons joined by `and`.

Unit 3

Comparison operator

An operator that compares two values and returns a boolean: `==`, `!=`, `<`, `>`, `<=`, `>=`.

Unit 3

Logical operator

`and`, `or`, and `not`, which combine or invert boolean values.

Unit 3

Operator precedence

The fixed order in which operators are evaluated. Parentheses override it.

Unit 3

Short-circuit evaluation

Stopping a boolean expression as soon as its result is known, so later parts are never evaluated.

Unit 3

Truth table

A table listing every combination of inputs and the resulting output of a logic expression.

Unit 3

Truthy / falsy

Values that count as True or False in a condition. Falsy values include `0`, `""`, `[]`, and `None`.

Unit 3

XOR (exclusive or)

True when exactly one of two inputs is True. In Python, `a != b` on booleans.

Unit 3

Unit 4: Conditionals: Making Decisions

Block

A group of statements that run together, introduced by a colon and indented beneath a statement.

Unit 4

Branch

One path through a conditional. Each `if`, `elif`, and `else` defines a branch.

Unit 4

Condition

An expression that evaluates to True or False and decides whether a block runs.

Unit 4

Conditional expression

A one-line choice written `a if condition else b`.

Unit 4

elif

Short for "else if". Tests another condition only if the previous ones were False.

Unit 4

Indentation

Leading spaces that show which lines belong to a block. Python requires consistent indentation.

Unit 4

Nested conditional

A conditional placed inside another conditional's block.

Unit 4

Unit 5: Loops: Repeating Work

Accumulator

A variable that collects a running result, such as a total, across loop iterations.

Unit 5

break

A statement that exits the nearest loop immediately.

Unit 5

continue

A statement that skips the rest of the current iteration and moves to the next.

Unit 5

for loop

A loop that assigns each item of a sequence to a variable in turn.

Unit 5

Infinite loop

A loop whose condition never becomes False, so it never ends unless interrupted.

Unit 5

Iteration

One pass through the body of a loop.

Unit 5

Loop

A statement that repeats a block of code.

Unit 5

Off-by-one error

A bug caused by a range or index that is one too high or too low.

Unit 5

range()

A built-in that produces a sequence of integers from `start` up to but not including `stop`.

Unit 5

while loop

A loop that repeats while its condition is True. The condition is checked before each pass.

Unit 5

Unit 6: Functions

Argument

The actual value passed to a function when it is called.

Unit 6

Default argument

A parameter with a preset value, such as `rate=0.07`, used when the caller does not supply one.

Unit 6

Docstring

A string literal placed as the first line of a function body to document what it does.

Unit 6

Function

A named, reusable block of code, defined with `def` and run by calling it.

Unit 6

Global variable

A variable defined at the top level of a program, visible to the code that follows it.

Unit 6

Keyword argument

An argument passed by name, such as `greet(name="Sam")`.

Unit 6

Local variable

A variable that exists only inside the function where it was created.

Unit 6

None

Python's special value meaning "no value". Functions without a `return` produce it.

Unit 6

Parameter

A variable name in a function definition that receives a value when the function is called.

Unit 6

return

A statement that ends a function and sends a value back to the caller.

Unit 6

Unit 7: Lists

Aliasing

When two names refer to the same object, so a change through one name is visible through the other.

Unit 7

enumerate()

A built-in that gives each item of a sequence together with its index, for use in a loop.

Unit 7

Index

The position of an item in a list, starting at 0. Negative indexes count from the end.

Unit 7

IndexError

The error raised when an index is outside the range of a list.

Unit 7

List

An ordered, mutable collection of values written in square brackets.

Unit 7

List comprehension

A compact expression that builds a list from a loop, such as `[n * n for n in range(5)]`.

Unit 7

Method

A function attached to an object. `append`, `insert`, `pop`, and `sort` are list methods.

Unit 7

Mutable

A value that can be changed after it is created. Lists and dictionaries are mutable.

Unit 7

Slice

A new list made from a range of positions, written `items[start:stop]`.

Unit 7

Unit 8: Tuples and Sets

Difference

The items in the first set that are not in the second. Written `a - b`.

Unit 8

Immutable

Cannot be changed after creation. Tuples, strings, and numbers are immutable.

Unit 8

Intersection

The set of items in both sets. Written `a & b`.

Unit 8

Membership test

Checking whether a value is in a collection with the `in` operator.

Unit 8

Set

An unordered collection of unique values, written in curly braces or created with `set()`.

Unit 8

Tuple

An immutable, ordered collection written in parentheses, such as `(4, 7)`.

Unit 8

Union

The set of items in either of two sets. Written `a | b`.

Unit 8

Unpacking

Assigning each item of a sequence to a separate name in one statement, such as `x, y = point`.

Unit 8

Unit 9: Dictionaries

.get()

A dictionary method that returns the value for a key, or a default if the key is missing.

Unit 9

.items()

A dictionary method that yields key-value pairs, usually unpacked in a loop as `for k, v in d.items()`.

Unit 9

Counter pattern

Using a dictionary to count occurrences: `counts[item] = counts.get(item, 0) + 1`.

Unit 9

Dictionary

A collection that maps unique keys to values, written in curly braces with `key: value` pairs.

Unit 9

Key

The label used to look up a value in a dictionary. Keys must be immutable and unique.

Unit 9

KeyError

The error raised when reading a key that is not in the dictionary.

Unit 9

Nested dictionary

A dictionary that contains another dictionary or list as a value.

Unit 9

Value

The data stored under a key. Values can be any type.

Unit 9

Unit 10: Libraries and Modules

__name__

A built-in variable. It equals `"__main__"` when a file is run directly, and the module name when imported.

Unit 10

Alias

A shorter name given to a module on import, such as `import numpy as np`.

Unit 10

import

A statement that loads a module so its names can be used.

Unit 10

Library

A collection of modules that provides a set of related tools, such as the standard library, NumPy, or pandas.

Unit 10

Module

A Python file (`.py`) whose functions and variables can be imported into another program.

Unit 10

pip

The package installer for Python. It downloads third-party libraries from the Python Package Index.

Unit 10

Seed

A starting value for a random number generator. The same seed produces the same sequence every run.

Unit 10

Standard library

The modules that come with every Python installation, such as `math`, `random`, and `json`.

Unit 10

A–Z

A

Accumulator · Alias · Aliasing · Argument · Assignment

B

Block · bool · Boolean · Branch · break

C

Casting · Chained comparison · Comment · Comparison operator · Condition · Conditional expression · continue · Counter pattern

D

Data type · Default argument · Dictionary · Difference · Docstring

E

elif · enumerate()

F

f-string · float · for loop · Function

G

Global variable

I

Immutable · Immutable · import · Indentation · Index · Index · IndexError · Infinite loop · int · Interpreter · Intersection · Iteration

K

Key · KeyError · Keyword · Keyword argument

L

Library · List · List comprehension · Local variable · Logical operator · Loop

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Membership test · Method · Method · Module · Mutable

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Truth table · Truthy / falsy · Tuple · type()

U

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Value · Variable

W

while loop

X

XOR (exclusive or)